

Notes: Fan and Yang (JPE, 2025)

Estimating Discrete Games with Many Firms and Many Decisions: An Application to Merger and Product Variety

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1 Big picture

- **Question.** How can we estimate discrete games when there are many firms or many binary decisions per firm, and what do such games imply for merger effects on entry, product variety, prices, and welfare?
- **Setting.** The paper develops a new estimation method for large discrete games and applies it to the California retail craft beer market, where firms are multiproduct breweries choosing which beer products to sell in each retailer-county market.
- **Core challenge.** Standard discrete-game methods become computationally prohibitive when the action space is large. In a product-choice game, each firm chooses a subset of potential products, so the number of possible outcomes grows exponentially in the number of firms and products.
- **Main methodological result.** The paper estimates the game using *bounds on conditional choice probabilities*. The lower bound is the probability that an action is dominant; the upper bound is the probability that the action is not dominated. These bounds are easy to compute and do not require solving for every equilibrium outcome.
- **Why this matters.** The method is valid under multiple equilibria, under market-specific equilibrium selection, and even when some markets have no pure-strategy equilibrium. This makes it much more scalable than approaches based on enumerating all possible equilibria.
- **Main empirical question.** What happens if a large macrobrewer acquires major craft brewers? Does entry by new firms and product adjustment by incumbents offset the merger’s negative price effects?
- **Main empirical results.**
 - The merger induces **new firm entry** in all markets.
 - Without fixed-cost merger efficiencies, the merged firm drops products; nonmerging incumbents and entrants add products, but not enough to prevent a fall in consumer surplus.
 - With fixed-cost efficiencies, the merged firm adds products, so total product variety can increase, but the merger still reduces consumer surplus and total surplus.
 - Thus, entry and variety adjustments mitigate but do not reverse the welfare loss from the merger.
- **Economic takeaway.** Product variety is an endogenous strategic margin. A merger can induce entry and yet still lower welfare because prices rise and incumbent product adjustments can be strategically substitutive.

2 Data and measurement (key objects)

2.1 Observed market-level and household-level data

- **Primary sources.** The main datasets are NielsenIQ Retail Scanner Data and NielsenIQ Consumer Panel data from 2010 to 2016.
- **Product definition.** A product is a beer brand in the NielsenIQ data, aggregated from UPC-week observations to product-month observations.

- **Quantity unit.** Quantities are homogenized into a *12-ounce 12-pack equivalent*.
- **Prices.** Product-month prices are quantity-weighted average prices across weeks within a month.
- **Supplementary data.** The authors add Brewers Association craft designations, hand-collected owner and brewery identities, production-facility locations, and county demographics from the US Census.

2.2 Markets, firms, products, and market size

- **Market definition.** A market is a *retailer-county pair*. This is motivated by consumer-panel evidence that cross-retailer shopping is rare: more than 80% of households in 2016 buy all their beer from one retailer-county combination.
- **Firm definition.** A firm is the corporate owner of a brand. One firm can own multiple breweries and products.
- **Availability rule.** A product is available in a market-year if monthly sales exceed 20 units for more than 6 months in that year.
- **Craft sample.** For craft products, the paper keeps beers produced by the top 60 craft breweries in the 2015 Brewers Association production data. The sample covers 91% of California craft-beer quantity in the scanner data over the sample period.
- **Potential product set.** A firm’s potential products in a year are all products it owns that are available in any market that year. The paper studies entry of *existing products into markets*, not new brewery creation or new brand creation.
- **Market size.** Market size is defined as average monthly alcohol sales in a market times 8, where 8 is the average number of household trips per month in the consumer panel.

2.3 Key descriptive objects

- In the balanced set of 110 markets present in all years, average annual craft quantity is about 4.71 million units, versus 53.16 million units for all beer.
- Average price is about \$17 for craft beer and \$11 for beer overall.
- On average, the data contain 35 craft firms and 129 craft products per year, versus 54 firms and 263 products overall.
- Craft beer is only about 10% of beer quantity, but craft firms and products make up more than half of the market by count.
- Among craft products, ales dominate: they account for about 69% of quantity and 66% of craft products. Light beer accounts for only 0.71% of craft quantity, despite being 41% of total beer quantity overall.
- In 2016, the fixed-cost estimation sample contains 95 potential craft products, 149 markets, and 14,155 potential product-market combinations.

2.4 Unobservables and timing

- **Demand shocks.** Consumer demand includes product-market-month shocks ξ_{jmt} , after controlling for product, market, and time fixed effects.
- **Marginal cost shocks.** Costs include product-market-month shocks ω_{jmt} .
- **Fixed-cost shocks.** Product-entry fixed costs contain an unobserved shock ζ_{jmt} , assumed i.i.d. normal in the baseline specification.
- **Timing.** Firms observe fixed costs when choosing annual product sets, but do *not* observe monthly demand and marginal-cost shocks at that stage. These shocks are realized later, before monthly pricing.

3 Models and empirical specifications (all equations + interpretation)

3.1 Illustrative 2×2 entry game and the basic bounds

The paper starts from a simple two-firm binary entry game. Firm n enters market m if postentry profit is nonnegative:

$$Y_{nm} = \mathbf{1}[\pi_{nm}(Y_{-nm}) - C_{nm} - \zeta_{nm} \geq 0]. \quad (1)$$

- $\pi_{nm}(Y_{-nm})$ is variable profit as a function of rival actions.
- C_{nm} is mean fixed cost.
- ζ_{nm} is an unobserved fixed-cost shock with distribution F_ζ .

The core behavioral assumption is **level-1 rationality**: the observed action is not dominated. Under this assumption,

$$\Pr(Y_{nm} = 1 \text{ is a dominant strategy}) \leq \Pr(Y_{nm} = 1) \leq \Pr(Y_{nm} = 1 \text{ is not a dominated strategy}). \quad (2)$$

If entry of rivals reduces profit, these bounds reduce to

$$F_\zeta(\pi_{nm}(1) - C_{nm}) \leq \Pr(Y_{nm} = 1) \leq F_\zeta(\pi_{nm}(0) - C_{nm}).$$

Interpretation. The lower bound says: if entry is profitable even in the worst rival-action configuration, then entry is dominant. The upper bound says: if entry is not ruled out even in the best rival-action configuration, then it is not dominated. These are much easier to compute than outcome-level equilibrium probabilities.

3.2 General model with many firms and many product decisions

There are N firms and M markets. Firm n chooses a vector of binary actions

$$Y_{nm} = (Y_{jm}, j \in J_n),$$

where J_n is firm n 's set of potential products.

The firm's profit is variable profit minus product-specific fixed costs:

$$\pi_n(Y_{nm}, Y_{-nm}, X_{nm}) - \sum_{j \in J_n} Y_{jm} (W_{jm} v + \sigma_\zeta \zeta_{jm}).$$

For a product j , define the change in variable profit from setting $Y_{jm} = 1$ rather than 0:

$$\Delta_j(Y_{-jm}, X_{nm}) = \pi_n(Y_{jm} = 1, Y_{-jm}, X_{nm}) - \pi_n(Y_{jm} = 0, Y_{-jm}, X_{nm}). \quad (4)$$

Then define its minimum and maximum over rivals' actions:

$$\underline{\Delta}_j(X_{nm}) = \min_{Y_{-jm}} \Delta_j(Y_{-jm}, X_{nm}), \quad \overline{\Delta}_j(X_{nm}) = \max_{Y_{-jm}} \Delta_j(Y_{-jm}, X_{nm}).$$

The conditional choice probability is bounded by

$$F_\zeta(\underline{\Delta}_j(X_{nm}) - c(W_{jm}, v), \theta_\zeta) \leq \Pr(Y_{jm} = 1 \mid X_{nm}, W_{jm}) \leq F_\zeta(\overline{\Delta}_j(X_{nm}) - c(W_{jm}, v), \theta_\zeta). \quad (5)$$

Define lower- and upper-bound moment functions:

$$L(Y_{jm}, X_{nm}, W_{jm}, v, \theta_\zeta) = F_\zeta(\underline{\Delta}_j(X_{nm}) - c(W_{jm}, v), \theta_\zeta) - \mathbf{1}[Y_{jm} = 1], \quad (6a)$$

$$H(Y_{jm}, X_{nm}, W_{jm}, v, \theta_\zeta) = \mathbf{1}[Y_{jm} = 1] - F_\zeta(\overline{\Delta}_j(X_{nm}) - c(W_{jm}, v), \theta_\zeta). \quad (6b)$$

These imply conditional moment inequalities:

$$\mathbb{E}[L(\cdot) \mid X_{nm}, W_{jm}] \leq 0, \quad \mathbb{E}[H(\cdot) \mid X_{nm}, W_{jm}] \leq 0. \quad (7)$$

The paper transforms them into unconditional moment inequalities using nonnegative instrument functions $g^{(k)}(X_{nm}, W_{jm})$:

$$\mathbb{E} \left[\frac{1}{\#J} \sum_{j \in J} L(Y_{jm}, X_{nm}, W_{jm}, v, \theta_\zeta) g^{(k)}(X_{nm}, W_{jm}) \right] \leq 0, \quad (8a)$$

$$\mathbb{E} \left[\frac{1}{\#J} \sum_{j \in J} H(Y_{jm}, X_{nm}, W_{jm}, v, \theta_\zeta) g^{(k)}(X_{nm}, W_{jm}) \right] \leq 0. \quad (8b)$$

Interpretation. The parameters (v, θ_ζ) are identified from how observed product-entry probabilities covary with exogenous profit shifters X_{nm} and fixed-cost shifters W_{jm} . The method estimates the structural distribution of entry shocks rather than only reduced-form deviations.

3.3 Approximation that makes the method scalable

Directly solving for $\underline{\Delta}_j$ and $\overline{\Delta}_j$ by enumerating all rival-action profiles is infeasible in large games. The paper shows that in settings with substitutable products one can approximate

$$\underline{\Delta}_j(X_{nm}) \approx \Delta_j((1, \dots, 1), X_{nm}), \quad \overline{\Delta}_j(X_{nm}) \approx \Delta_j((0, \dots, 0), X_{nm}).$$

Interpretation. The worst case for adding a product is when all rivals' products are present; the best case is when none are. In the authors' computational checks, these approximations coincide with the exact extrema in all sampled cases.

3.4 Monte Carlo design

In the Monte Carlo exercise, firms play a binary entry game with variable profit

$$\pi_n(Y_{-nm}, X_{nm}) = \Omega_m x_{nm} - \phi \log \left(1 + \sum_{n' \neq n} z_{n'm} Y_{n'm} \right),$$

and fixed cost $C + \sigma_\zeta \zeta_{nm}$, with true parameters $C = 1$ and $\sigma_\zeta = 1$.

Interpretation. The exercise shows two things. First, the confidence set becomes tighter around the true parameter values. Second, the computational advantage of the Fan-Yang bounds becomes enormous as the number of firms rises.

3.5 Demand system for beer

Household i 's utility from product j in market m and month t is

$$\begin{aligned} u_{ijmt} = & (\sigma_0 \nu_i + \kappa_0 y_i) + (\alpha + \kappa_a y_i) p_{jmt} + \sigma^{ale} \nu_i^{ale} x_j^{ale} + \sigma^{lager} \nu_i^{lager} x_j^{lager} + \sigma^{light} \nu_i^{light} x_j^{light} \\ & + \sigma^{import} \nu_i^{import} x_j^{import} + (\sigma^{craft} \nu_i^{craft} + \kappa^{craft} y_i) x_j^{craft} + d_{jm} \beta + F E_j^{demand} + F E_m^{demand} + F E_t^{demand} \\ & + F E_m^{demand, craft} + F E_t^{demand, craft} + \xi_{jmt} + \varepsilon_{ijmt}. \end{aligned} \quad (12)$$

- Product characteristics are flavor type, import status, and craft designation.
- Household income y_i affects both overall beer demand and price sensitivity.
- Unobserved taste heterogeneity ν_i follows a normal distribution.
- The idiosyncratic error ε_{ijmt} is i.i.d. type-I extreme value.
- Distance between brewery and market enters only for craft beers through $d_{jm} \beta$.

This yields market shares $s_{jmt}(p_{jmt}, p_{-jmt})$ and thus market demand

$$D_{jmt}(p_{jmt}, p_{-jmt}) = s_{jmt}(p_{jmt}, p_{-jmt}) \times \text{market size}.$$

Interpretation. The demand system is deliberately flexible because craft beer is highly differentiated and household taste heterogeneity is central to the product-variety counterfactuals.

3.6 Supply side: two-stage game

The supply side is a two-stage static game.

Stage 2: pricing. Conditional on the chosen product set, firm n chooses monthly prices for all products $j \in J_{nmt}$ to maximize variable profit:

$$\max_{p_{jmt}, j \in J_{nmt}} \sum_{j \in J_{nmt}} (p_{jmt} - mc_{jmt}) D_{jmt}(p_{jmt}, p_{-jmt}). \quad (13)$$

Marginal cost is specified as

$$mc_{jmt} = F E_j^{mc} + F E_m^{mc} + F E_t^{mc} + F E_m^{mc, craft} + F E_t^{mc, craft} + d_{jm} \gamma + \omega_{jmt}. \quad (14)$$

Interpretation. Markups can vary by product, market, and month through the fixed effects. Distance also shifts marginal cost via transportation or distribution costs.

Stage 1: entry and product choice. At the beginning of year t , firm n chooses a subset of potential products J_{nt} to offer in market m :

$$\max_{J_{nmt} \subseteq J_{nt}} \pi_{nm}(J_{nmt}, J_{-nmt}) - C_{nm}(J_{nmt}). \quad (15)$$

Expected variable profit is the annual sum of expected monthly variable profits,

$$\pi_{nm}(J_{nmt}, J_{-nmt}) = \sum_{t \in T_\tau} \mathbb{E}_{\xi_{mt}, \omega_{mt}} \left(\sum_{j \in J_{nmt}} (p_{jmt}(J_{nmt}, J_{-nmt}) - mc_{jmt}) Q_{jmt}(J_{nmt}, J_{-nmt}) \right). \quad (16)$$

The fixed cost of offering the product set is

$$C_{nm}(J_{nmt}) = \sum_{j \in J_{nmt}} (W_{jm}\theta + \sigma_\zeta \zeta_{jmt}). \quad (17)$$

Interpretation. Firms pick product portfolios before transient demand and cost shocks are realized. Hence, selection at the product-choice stage occurs through fixed-cost shocks, while demand and marginal-cost heterogeneity is soaked up by rich fixed effects.

3.7 Demand estimation and marginal costs

Demand is estimated by GMM using macro moments plus two sets of micro moments.

- **Macro moments.** Prices are instrumented using interactions of global barley prices with beer types.
- **Micro moments from persistence.** The panel structure of household purchases identifies dispersion parameters in tastes. For example, if craft taste heterogeneity is large, households that ever buy craft beer in a year should buy a lot of it over the year conditional on ever buying it.
- **Micro moments from income bins.** These identify income effects on price sensitivity and craft demand.

Marginal costs are backed out using the first-order conditions from the pricing problem in equation (13), following standard BLP-style logic.

3.8 Fixed-cost estimation and identification in the application

The paper reformulates the product-choice problem in indicator form as

$$\pi_n(Y_{nm}, Y_{-nm}, X_{nm}) - \sum_{j \in J_n} Y_{jm}(W_{jm}\theta + \sigma_\zeta \zeta_{jm}). \quad (18)$$

Identification uses two main sources of exogenous variation:

- **Market size.** Larger markets raise variable profits and therefore entry probabilities.

- **Distance from brewery to market.** In-state craft products are much more likely to appear in nearby markets than faraway markets, conditional on being part of the potential product set.

Interpretation. The bounds are informative because many craft products have low diversion to noncraft products, which makes the gap between the lower and upper profit changes relatively small.

3.9 Counterfactual merger design

The paper studies a counterfactual in which the largest macrobrewer acquires the top three to five craft producers, pushing craft-market concentration closer to that of the overall beer market.

Three counterfactuals are computed:

- **CF1:** entry, incumbent product adjustment, and price adjustment all allowed.
- **CF2:** only incumbent product adjustment and price adjustment.
- **CF3:** only price adjustment, holding premerger product sets fixed.

The decomposition is:

- overall product-variety effect: $CF1 - CF3$,
- entry-driven variety effect: $CF1 - CF2$,
- incumbent-adjustment variety effect: $CF2 - CF3$.

A merger-efficiency scenario is also considered, where acquired craft products enjoy a reduction in fixed costs equal to the estimated extra fixed cost of independent craft breweries.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard discrete-game approach

- Maximum-likelihood approaches are difficult because multiple equilibria require explicit selection rules.
- Ciliberto–Tamer style bounds are sharper but computationally explode because they require constructing bounds for every possible market outcome.
- Necessary-condition moment inequalities are more tractable, but they usually do not estimate the structural distribution of entry shocks and therefore are less suitable for counterfactuals with structural fixed-cost heterogeneity.

4.2 What identifies the new approach

- The method uses **exogenous variation in variable profits** through X_{nm} , such as market size and distance.
- It uses **exogenous variation in fixed costs** through W_{jm} , such as craft status, California location, and market-size bins.

- It identifies both the **mean fixed-cost parameters** and the **distribution of fixed-cost shocks** by comparing how sharply entry probabilities respond to these shifters.

4.3 Why the bounds are useful here

- The craft-beer application is a product-portfolio game with many potential products per firm, so the full action space is enormous.
- The conditional-choice-probability bounds are one-dimensional and easy to compute.
- The model is especially informative because substitution between craft and noncraft products is weak, which keeps the lower and upper profit changes close together for many products.

4.4 Empirical credibility of the application

- The demand system has rich product, market, and time fixed effects, so remaining shocks are only transient product-market-month shocks.
- The R^2 's from regressing mean utility and marginal cost on observables and fixed effects are both above 0.9, implying that the residual transient shocks are small.
- The counterfactual merger is designed to preserve consistency with the observed premerger equilibrium by drawing fixed-cost shocks conditional on the observed outcome being an equilibrium.

4.5 Relation to the literature

- Methodologically, the paper extends the literature on incomplete discrete games and moment inequalities by using bounds built from *dominant* and *nondominated* actions.
- Empirically, it contributes to the merger-and-entry literature by studying *multiproduct firms with endogenous product choice* rather than only firm entry.
- It also contributes to the product-variety literature by showing that merger efficiencies in fixed costs can have countervailing effects: they may increase merged-firm product variety while reducing entry and rival product expansion.

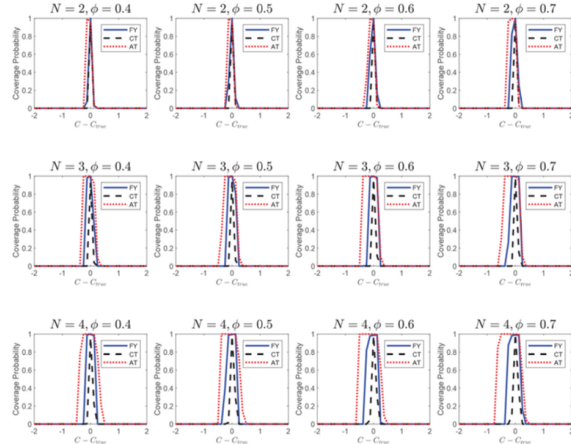
5 Tables and figures

5.1 Figure 1 and Table 1: Monte Carlo performance and computation

Read:

- The figure compares coverage properties of the Fan–Yang bounds with the Ciliberto–Tamer and Aradillas-Lopez–Tamer bounds.
- The main message from the Monte Carlo is not that the new bounds are always sharper; it is that they remain informative while being dramatically easier to compute.

A C



B σ

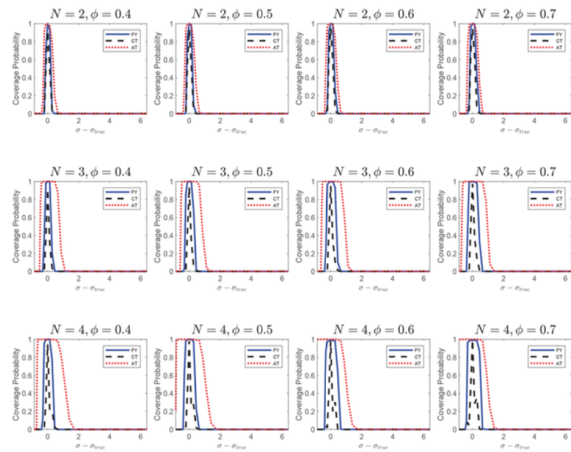


FIG. 1.—Coverage probabilities: FY versus CT versus AT bounds.

TABLE 1
COMPUTATION TIME: FY VERSUS CT VERSUS AT BOUNDS

N	FY (s) (1)	CT (s) (2)	AT (s) (3)
2	.017	.341	.017
3	.014	.739	.084
4	.022	1.930	.452
5	.023	5.580	1.682
6	.021	21.743	13.891
7	.016	151.646	132.086
8	.126	1,459.528	1,421.882

NOTE.—The table reports the computation time required to evaluate the test statistic used for constructing the confidence set once. N = number of potential entrants.

- Table 1 is the punchline for scalability: with 8 potential entrants, the Fan–Yang test statistic takes about 0.126 seconds to compute, while CT and AT methods take roughly 1,460 and 1,422 seconds, respectively.
- So the paper’s key methodological gain is computational feasibility in large games.

5.2 Table 2 and Table 3: Market background and beer composition

TABLE 2
ANNUAL TOTAL QUANTITY, PRICES, AND NUMBER OF FIRMS AND PRODUCTS

	Total Quantity (12-Ounce 12-Pack Equivalent)	Average Price (2016 US\$)	Firms per Year	Products per Year
Craft	4,710,634	17	35	129
All	53,155,825	11	54	263

NOTE.—For each year from 2010 to 2016, we first calculate a year’s total and craft beer quantities, quantity-weighted average prices, number of firms, and number of products and then take the average across years.

TABLE 3
SHARES OF TOTAL QUANTITY AND NUMBER
OF PRODUCTS BY BEER TYPES

	Ale	Lager	Light
Quantity (%):			
Craft	69.10	29.34	.71
All	11.9	46.07	40.94
Number of products (%):			
Craft	65.82	27.44	.50
All	43.11	40.38	7.60

NOTE.—Shares reflect the respective proportions of the total quantity from 2010 to 2016 or of the total number of unique products in these years.

Read:

- Table 2 shows that craft beer is a small share of beer quantity but a large share of firm and product counts. This is exactly the type of market where product-variety margins matter for merger analysis.
- Craft beer averages about \$17 per 12-pack-equivalent versus \$11 overall, suggesting higher prices and potential differentiation.
- Table 3 shows that craft is overwhelmingly an ale business, whereas light beer is almost absent inside craft. This matters later for the estimated negative taste correlation between craft and light beer.

5.3 Figure 3: Barley prices as instruments

Read:

- The figure shows meaningful time variation in global barley prices.
- The authors interact barley prices with beer-type indicators to instrument for retail prices in demand estimation.
- The idea is standard cost-shifter logic: barley moves supply-side costs, and therefore prices, without directly shifting consumer tastes.

5.4 Table 4 and Table 5: Demand estimates and micro moments

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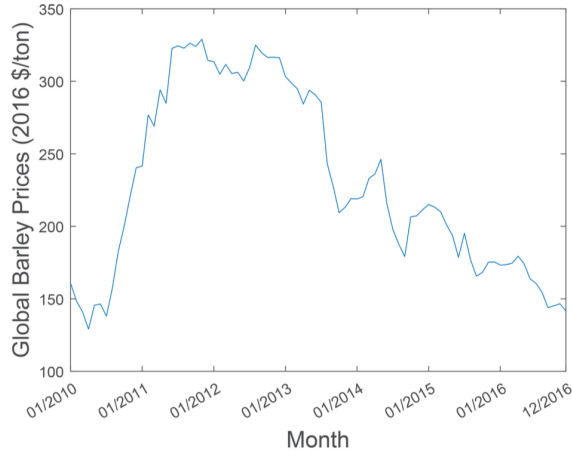


FIG. 3.—Price of barley.

TABLE 4
DEMAND ESTIMATES

Parameter	Estimate
Unobserved heterogeneity:	
σ_0	.001 (.01)
σ^{ale}	2.03 (<.01)
σ^{lager}	.86 (<.01)
σ^{light}	2.56 (<.01)
σ^{import}	1.82 (<.01)
σ^{craft}	2.45 (<.01)
$\rho^{\text{craft-light}}$	-.75 (<.01)
Income effect:	
κ_0	-1.34 (.02)
κ^{craft}	.84 (.02)
κ_α	.08 (<.01)
Price coefficient:	
α	-1.50 (.03)
Fixed effects:	
Product	Yes
Distance bin	Yes
Distance bin $\times$ craft	Yes
Market	Yes
Market $\times$ craft	Yes
Time	Yes
Time $\times$ craft	Yes

NOTE.—Standard errors are in parentheses.

TABLE 5
SELECTED MICRO MOMENTS ON PERSISTENCE IN
PURCHASING DECISIONS

	Data	Model
1. $E(\sum_{t=1}^{12} q_{it} \sum_{t=1}^{12} q_{it} \geq 1)$	7.50	7.72
2. $E(\sum_{t=1}^{12} q_{it}^{\text{ale}} \sum_{t=1}^{12} q_{it}^{\text{ale}} \geq 1)$	3.10	4.10
3. $E(\sum_{t=1}^{12} q_{it}^{\text{lager}} \sum_{t=1}^{12} q_{it}^{\text{lager}} \geq 1)$	5.56	4.07
4. $E(\sum_{t=1}^{12} q_{it}^{\text{light}} \sum_{t=1}^{12} q_{it}^{\text{light}} \geq 1)$	8.03	8.02
5. $E(\sum_{t=1}^{12} q_{it}^{\text{import}} \sum_{t=1}^{12} q_{it}^{\text{import}} \geq 1)$	2.86	2.57
6. $E(\sum_{t=1}^{12} q_{it}^{\text{craft}} \sum_{t=1}^{12} q_{it}^{\text{craft}} \geq 1)$	3.93	4.12

- Table 4 shows large unobserved heterogeneity in tastes for craft beer ($\hat{\sigma}^{craft} = 2.45$) and imported beer, as well as a strong negative correlation between craft and light tastes ($\hat{\rho}_{craft,light} = -0.75$).
- High-income households are less price sensitive and more likely to prefer craft beer.
- Table 5 shows the model matches persistence moments well. In particular, households that buy any craft beer in a year buy a lot of craft beer over the year, which is what identifies large craft-taste heterogeneity.

5.5 Table 6 and Figure 4: Elasticities, diversion ratios, and markups

TABLE 6
ELASTICITIES OF TOP THREE CRAFT PRODUCTS AND
TOP THREE MAIN PRODUCTS (%)

	Craft			Main		
	1	2	3	1	2	3
Craft:						
1	-10.05	.13	.02	.01	.01	.03
2	.21	-9.40	.02	.01	.01	.03
3	.04	.02	-9.07	.01	.01	.08
Main						
1	<.01	<.01	<.01	-5.25	.56	.04
2	<.01	<.01	<.01	.57	-5.25	.04
3	<.01	<.01	.01	.04	.04	-8.87

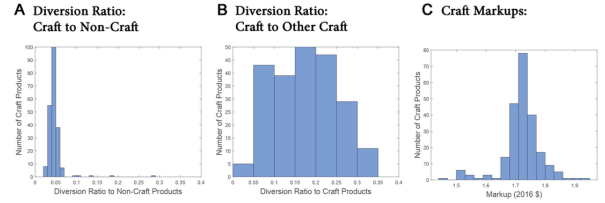


FIG. 4.—Histograms of diversion ratios and markups.

Read:

- Own-price elasticities are large in magnitude for both top craft and top mainstream beers. For the top craft products they are roughly -10 , -9.4 , and -9.1 .
- Cross-elasticities from craft to mainstream products are near zero, while diversion from one craft product to other craft products is substantial.
- Figure 4 shows almost no diversion from a craft price increase toward noncraft products, but meaningful diversion toward other craft products.
- The median craft markup is about \$1.7, which is economically reasonable relative to industry benchmarks.

5.6 Figure 5 and Figure 6: Identification in the application

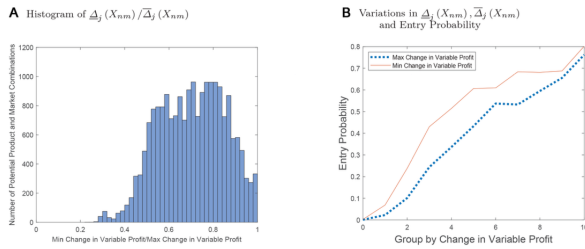


FIG. 5.—Data patterns aiding identification.

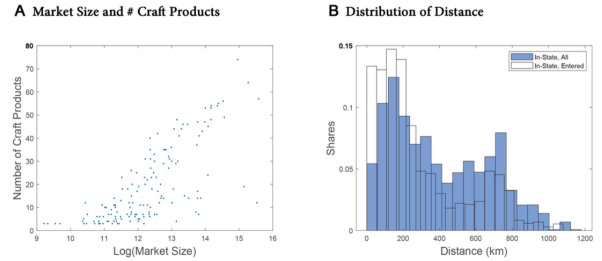


FIG. 6.—Exogenous variations aiding identification.

Read:

TABLE 7
ESTIMATES OF FIXED COSTS: PROJECTED 95%
CONFIDENCE INTERVALS (2016)

	Projected 95% Confidence Interval
Craft (θ_1)	255.32 to 583.95
In state $\times$ craft (θ_2)	-276.04 to 37.57
Market size-specific fixed cost (θ_3):	
Small market	426.52 to 721.90
Medium market	720.77 to 1,014.10
Large market	2,028.00 to 2,453.69
Market size-specific SD (σ_i):	
Small market	.02 to 105.80
Medium market	1.18 to 207.45
Large market	767.59 to 1,044.93

NOTE.—Estimates are in 2016 US dollars.

- Figure 5 shows that lower and upper profit changes are often relatively close, which means the conditional choice probability bounds are fairly tight for many observations.
- Entry probabilities increase strongly with the profit-change statistics, exactly as the identification argument requires.
- Figure 6 shows the two key exogenous shifters: market size and distance. Larger markets support more craft products, and products are more likely to be present when the brewery is closer to the market.

5.7 Table 7: Fixed-cost estimates

Read:

- Independent craft breweries face higher fixed costs than noncraft products: the projected 95% confidence interval for the craft coefficient is \$255 to \$584 in 2016 dollars.
- Fixed costs rise strongly with market size: about \$427–\$722 in small markets, \$721–\$1,014 in medium markets, and \$2,028–\$2,454 in large markets.
- The standard deviation of fixed-cost shocks also rises with market size, meaning large markets are not only more expensive but also more uncertain.

5.8 Figure 7 and Table 8: Merger effects on entry, variety, prices, and welfare

Read:

- Without merger efficiency, the merger induces both entry and exit, with more entries than exits. New entrants bring new products, and nonmerging incumbents also add products, but the merged firm drops products.
- The quantity-weighted average craft-beer price increase is centered around 5 cents and can be as large as 15 cents, about 10% of the average markup.
- Total surplus falls by roughly \$200,000 to \$780,000, while consumer surplus falls by about \$233,000 to \$1.075 million. Craft-beer profits rise.
- With merger efficiency, the merged firm adds products and total product count can rise, but total surplus is still negative (about \$31,000 to \$402,000 lower) and consumer surplus still falls (about \$152,000 to \$670,000 lower).

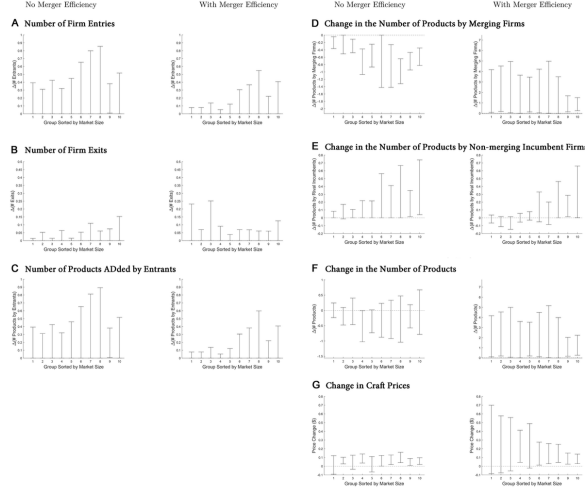


FIG. 7.—Merger effects on entry, product variety, and prices.

TABLE 8
WELFARE EFFECTS

	No Merger Efficiency (1)	With Merger Efficiency (2)
1. Total surplus (US\$1,000s)	−780.05 to −200.23	−401.66 to −31.24
2. Craft beer profits (US\$1,000s)	33.02 to 294.99	25.34 to 285.68
3. Consumer surplus (US\$1,000s)	−1,075.04 to −233.25	−670.21 to −152.21
4. Due to variety change	−91.00 to 111.55	6.08 to 566.06
5. Due to entry	1.21 to 122.86	1.14 to 69.95
6. Due to incumbent product adjustments	−111.05 to −9.11	4.94 to 519.47

NOTE.—The table reports the aggregate welfare effects of the merger across markets. For each measure, we report the range across the vectors of parameters sampled from their 95% confidence set. Column 1 shows the results without considering any merger efficiency, while col. 2 reports the results that incorporate reductions in fixed costs when a craft brewery is acquired by a macrobrewery.

- The key substantive message is that new entry and product variety mitigate the welfare loss but do not offset the higher prices created by the merger.

6 Short summary

- **Method.** The paper develops a scalable estimator for large discrete games using bounds on conditional choice probabilities derived from dominant and nondominated strategies.
- **Identification.** The method uses exogenous variation in variable profits and fixed costs to identify both mean fixed-cost parameters and the distribution of fixed-cost shocks.
- **Application.** In California craft beer, demand is highly differentiated within craft and weakly substitutable between craft and noncraft. Independent craft breweries face higher entry fixed costs, and these costs rise with market size.
- **Merger result.** A macrobrewer's acquisition of major craft firms induces entry, product adjustment, and higher prices. Merger efficiencies in fixed costs can increase total product variety, but they still do not eliminate the negative effect of the merger on consumer welfare.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that large discrete games can be estimated without solving for all equilibria or imposing an equilibrium-selection rule.
- It also shows that merger analysis can be seriously incomplete if it treats product variety and entry as fixed. Multiproduct incumbents respond strategically, and those responses matter for welfare.
- In the craft-beer application, merger-induced entry is real, but it is not enough to overturn the merger's negative price effects.

7.2 Economic intuition

- The method works because dominance and nondominance are much easier to evaluate than full equilibrium outcome probabilities.
- On the economic side, product variety is an equilibrium object. When one firm merges and changes its product set, both incumbents and potential entrants react.
- Fixed-cost efficiencies do not mechanically raise welfare. They can encourage the merged firm to add products, but they also soften incentives for rivals and new firms to enter or expand because product choices are strategic substitutes.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it provides a practical way to estimate rich discrete games with many firms and many decisions. Substantively, it shows that merger analysis in differentiated-product industries should account jointly for prices, entry, and multiproduct variety choices. In the California craft-beer market, these margins all move after a merger, but the final welfare conclusion remains clear: even with favorable entry conditions and even with fixed-cost efficiencies, the merger lowers consumer welfare.